

Experiment 1

Name-Arya Gupta
Reg no.-25BDS0279

Program 1:

Finding a factorial upto a given number

```
num=int(input("Enter a number. "))
factorial=1
if num<0:
    print("Not defined")
else:
    for i in range(1,num+1):
        factorial*=i
print(f"Factorial of {i} is {factorial}")
```

```
===== RESTART: /home/matlab/25BDS0279.py =====
Enter a number: 5
Factorial of 1 is 1
Factorial of 2 is 2
Factorial of 3 is 6
Factorial of 4 is 24
Factorial of 5 is 120
```

Program 2:

Check if a number is
Armstrong or not.

```
num=int(input("Enter a number:
")) temp=num
sum=0
digits=len(str(num))
while temp>0:
    digit=temp%10
    sum+=digit**dig
    its temp//=10
if sum==num:
    print(num, "is an Armstrong
number.") else:
print(num, "is not an Armstrong number")
```

```
===== RESTART: /home/matlab/25BDS0279.py =====
```

```
Enter a number: 153  
153 is an Armstrong number.
```

```
===== RESTART: /home/matlab/25BDS0279.py =====
```

```
Enter a number: 235  
235 is not an Armstrong number.
```

Program 3: Removing duplicates from a list

```
numbers=[]  
n=int(input("Enter the number of  
elements: "))  
for i in range(n):  
    num=int(input("Enter  
elements" ))  
    numbers.append(num)  
unique=[]  
for item in numbers:  
    if item not in unique:  
        unique.append(item)  
print("Original List:", numbers)  
print("List after removing duplicates:", unique)
```

```
===== RESTART: /home/matlab/25BDS0279.py =====
```

```
Enter the number of elements: 7  
Enter element: 10  
Enter element: 20  
Enter element: 20  
Enter element: 50  
Enter element: 80  
Enter element: 10  
Enter element: 60  
Original List: [10, 20, 20, 50, 80, 10, 60]  
List after removing duplicates: [10, 20, 50, 80, 60]
```

Program 4: Frequency of numbers in a list

```
numbers=[1,2,3,4,3,5,2,1,2,4,4]  
frequency={}  
for num in numbers:  
    if num in frequency:  
        frequency[num]+=1  
    else:  
        frequency[num]=1  
print(frequency)
```

```
===== RESTART: /home/matlab/25BDS0279.py =====
```

```
{1: 2, 2: 3, 3: 2, 4: 3, 5: 1}
```

Program 5:

Length of words in a sentence

```
sentence=input("Enter a  
sentence: ")
```

```
words=sentence.split()
```

```
length={}
```

```
for word in words:
```

```
    length[word]=len(word)
```

```
print(length)
```

```
===== RESTART: /home/matlab/25BDS0279.py =====  
Enter a sentence: I am studying in VIT Vellore.  
{'I': 1, 'am': 2, 'studying': 8, 'in': 2, 'VIT': 3, 'Vellore.': 8}
```

Program 6:

Employee Salary Management

```
employee = {}
```

```
n = int(input("Enter number of employees: "))
```

```
for i in range(n):
```

```
    name = input("Enter employee name: ")
```

```
    salary = int(input("Enter salary: "))
```

```
    employee[name] = salary
```

```
highest = max(employee, key=employee.get)
```

```
print("\nEmployee Details")
```

```
for name, salary in employee.items():
```

```
    print(name, salary)
```

```
print("\nHighest Paid Employee")
```

```
print(highest, employee[highest])
```

```
C: > Users > Arya > Desktop > expq6.py > ...  
1   employee = {}  
2  
3   n = int(input("Enter number of employees: "))  
4  
5   for i in range(n):  
6       name = input("Enter employee name: ")  
7       salary = int(input("Enter salary: "))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Enter employee name: Shristi  
Enter salary: 70000  
Enter employee name: Sneha  
Enter salary: 65000
```

Employee Details

```
Arya 50000  
Aryan 45000  
Shristi 70000  
Sneha 65000
```

Highest Paid Employee

```
Shristi 70000
```

```
PS C:\Users\Arya\Desktop\webdev> █
```

Program 7:

Password Strength Checker

```
password = input("Enter password: ")
upper = lower = digit = special = 0
for ch in password:
    if ch.isupper():
        upper += 1
    elif ch.islower():
        lower += 1
    elif ch.isdigit():
        digit += 1
    else:
        special += 1
if len(password) >= 8 and upper > 0 and lower > 0 and digit > 0 and special > 0:
    print("Strong Password")
else:
    print("Weak Password")
```

```
PS C:\Users\Arya\Desktop\webdev> & C:/Users/Arya/AppData/Local/Programs/Python/Python314/python.exe c:/Users/Arya/Desktop/expq6.py
Enter password: Arya##1234
Strong Password
PS C:\Users\Arya\Desktop\webdev> & C:/Users/Arya/AppData/Local/Programs/Python/Python314/python.exe c:/Users/Arya/Desktop/expq6.py
Enter password: arya12
Weak Password
PS C:\Users\Arya\Desktop\webdev> █
```

Program 8:

Convert a Tuple into a Dictionary

```
keys = ("Name", "Age", "City")
values = ("Arya", 19, "Mumbai")
result = dict(zip(keys, values))
print(result)
```

```
PS C:\Users\Arya\Desktop\webdev> & C:/Users/Arya/AppData/Local/Programs/Python/Python314/python.exe c:/Users/Arya/Desktop/expq6.py
{'Name': 'Arya', 'Age': 19, 'City': 'Mumbai'}
PS C:\Users\Arya\Desktop\webdev> █
```

Program 9:

Compress a String

```
s = input("Enter a string: ")
count = 1
result = ""
for i in range(len(s)):
    if i < len(s) - 1 and s[i] == s[i + 1]:
        count += 1
    else:
        result += s[i] + str(count)
        count = 1
print("Compressed String:", result)
```

```
PS C:\Users\Arya\Desktop\webdev> & C:/Users/Arya/AppData/Local/Programs/Python/Python314/python.exe c:/Users/Arya/Desktop/expq6.py
Enter a string: bdbffeee
Compressed String: b1d1b1f2e3
PS C:\Users\Arya\Desktop\webdev>
```

Program 10:

Find Union, Intersection, and Difference of Two Sets

```
set1 = set(map(int, input("Enter elements of Set 1: ").split()))
set2 = set(map(int, input("Enter elements of Set 2: ").split()))
print("Union:", set1 | set2)
print("Intersection:", set1 & set2)
print("Difference (Set1 - Set2):", set1 - set2)
print("Difference (Set2 - Set1):", set2 - set1)
```

```
PS C:\Users\Arya\Desktop\webdev> & C:/Users/Arya/AppData/Local/Programs/Python/Python314/python.exe c:/Users/Arya/Desktop/expq6.py
Enter elements of Set 1: 1 3 5 6 9
Enter elements of Set 2: 2 3 4 5 6
Union: {1, 2, 3, 4, 5, 6, 9}
Intersection: {3, 5, 6}
Difference (Set1 - Set2): {1, 9}
Difference (Set2 - Set1): {2, 4}
PS C:\Users\Arya\Desktop\webdev>
```